



PUBLIC WHITE PAPER | WP-006

Flagship Pilot Design and Commercial Validation

How to turn one customer deployment into defensible operational, economic and market evidence

PURPOSE

Define a disciplined Canadian flagship-pilot framework for selecting the right customer, locking a credible baseline, controlling technical and regulatory risk, measuring value, and deciding whether the evidence supports purchase, repetition, revision or stop.

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Executive summary

A flagship pilot is not valuable because equipment is visible at a customer site. It is valuable only when the deployment answers a defined commercial question with traceable data, a controlled configuration and agreed decision rules.

Rapid Organic should treat its first Canadian RO-50 pilot as a commercial experiment. The site must have a material waste problem, measurable baseline costs, an accountable sponsor, compatible feedstock, suitable utilities, a lawful output route and willingness to share the evidence needed for a credible conclusion.

Pilot objective	Evidence required	Decision enabled
Operational value	Measured changes in storage, handling, hauling, disposal, labour or other agreed pain points.	Approve a bounded value claim.
Real-world usability	Training, loading, cleaning, interventions and operator observations.	Define staffing, SOPs and training.
Site reliability	Availability, failures, alarms, maintenance and response records.	Set service, spares and warranty assumptions.
Customer economics	Signed baseline, pilot costs, residual handling and sensitivity analysis.	Approve an offer and ROI range.
Adoption readiness	Sponsor, operator acceptance, site fit and continuation decision.	Proceed, revise, narrow or stop.
Publishable proof	Approved case study, data package, limitations and permissions.	Support sales and strategic outreach.

CORE DECISION

The flagship pilot must not begin until the baseline, configuration, success criteria, data rights, output route, responsibilities and stop rules are written and accepted.

1. What the pilot can—and cannot—prove

Within scope

- One named customer problem at one Canadian site.
- One frozen RO-50 reference configuration.
- One approved feedstock envelope and operating procedure.
- One defined baseline period and representative pilot window.
- Measured technical, operating and economic outcomes within those boundaries.

Outside scope

- Universal feedstock compatibility or performance in every season and site.
- RO-100 capacity or scale-up performance.
- Automatic fertilizer, compost, pathogen-free or unrestricted land-application status.
- Guaranteed economics for customers with different hauling, labour or utility costs.
- Commercial scalability before service, manufacturing and repeat-sale evidence exist.

EVIDENCE BOUNDARY

A successful pilot supports a bounded claim about the tested use case. It does not convert one result into a universal product claim.

2. Pilot stage-gate architecture

Gate	Purpose	Mandatory exit package
CP0 — Qualify	Confirm a real and measurable customer problem.	Opportunity brief, waste profile, sponsor, data availability and regulatory screen.
CP1 — Design	Define the value hypothesis and boundaries.	Pilot charter, KPI tree, baseline plan, responsibilities, budget and risk register.
CP2 — Contract	Make the deployment executable and controlled.	Signed agreement, site survey, insurance, utilities, output route, installation and training plans.
CP3 — Lock baseline	Approve what “before” means.	Signed baseline dataset, assumptions, calculation rules, exclusions and evidence rating.
CP4 — Operate	Generate representative evidence.	Accepted operating window, logs, deviations, service records and sampling results.
CP5 — Evaluate	Determine technical and customer value.	Final report, economics, limitations, customer acceptance and decision memo.
CP6 — Convert	Control what happens next.	Purchase, paid extension, repeat-pilot, redesign or stop decision; approved publication package.

A gate is not a calendar milestone. Each gate closes only when its evidence package is complete and the named decision authority records a decision.

3. Selecting the flagship customer

Criterion	Strong signal	Warning signal
Problem value	Material, recurring and already budgeted waste cost or operational pain.	Interest is mainly publicity or curiosity.
Feedstock fit	Source-separated, characterized and within the approved test envelope.	Highly mixed, seasonal or frequently changing material.
Data quality	Invoices, weights, pickups, labour and incident records are available.	Savings must be reconstructed from memory.
Sponsor	Named manager with authority, time and a reason to succeed.	No accountable owner or frequent internal handoffs.
Site readiness	Utilities, space, access, drainage, safety and output storage can be controlled.	Major unresolved building or environmental work.
Commercial potential	Plausible purchase path and repeatable segment.	Free trial with no decision process or budget.
Evidence rights	Customer accepts reasonable data access and bounded publication.	All results must remain inaccessible indefinitely.

RECOMMENDED PROFILE

Prefer a customer with a painful but manageable waste stream, clean baseline data and a committed operating team—not necessarily the largest or most recognizable organization.

4. The pilot charter

Charter element	Required content
Decision question	The exact question management expects the pilot to answer.
Use case	Site, feedstock, current practice, pain point and proposed intervention.
Boundaries	Machine revision, feedstock envelope, operating window, exclusions and output route.
Hypothesis	Expected operational or economic change and the causal pathway.
KPIs	Definition, source, owner, frequency, acceptance band and calculation rule.
Responsibilities	Customer and Rapid Organic obligations, staffing and escalation.
Schedule and budget	Baseline, installation, commissioning, operation, evaluation and removal/transfer.
Decision outcomes	Buy, extend, repeat, revise or stop; no automatic conversion.

The charter should be short enough to use and specific enough to prevent the pilot from changing meaning after results appear.

5. Baseline before benefit

Baseline domain	Minimum evidence
Waste generation	Weights or defensible volume-to-mass method by day, source and material type.
Current handling	Bins, refrigeration, storage time, internal movements, cleaning and labour.
Hauling/disposal	Invoices, pickup frequency, surcharges, contamination fees and contract terms.
Incidents and nuisance	Odor, pests, leakage, complaints, shutdowns or rejected loads with documented sources.
Utilities and space	Power, water, drainage, floor area and other opportunity costs relevant to comparison.
Output value	Zero unless a lawful receiver, quality specification and defensible value exist.
Seasonality	Production peaks, shutdowns, temperature and other variables affecting comparison.

BASELINE RULE

If the customer and Rapid Organic cannot agree on the “before” data and calculation rules, they will not agree on the savings after the pilot.

6. KPI governance

KPI	Definition control	Typical evidence source
Accepted input	Eligible material actually loaded; rejects reported separately.	Calibrated scale and batch record.
Processed output	Wet and dry quantities by defined output stream.	Scale, sampling and laboratory record.
Availability	Scheduled operating time capable of performing intended function.	State log plus reviewed downtime coding.
Intervention	Unplanned operator or service actions outside standard work.	Operator and service logs.
Energy intensity	Metered machine energy divided by accepted input on stated basis.	Dedicated meter and scale.
Cost avoided	Only baseline cost demonstrably displaced during accepted window.	Invoices, contract rules and approved model.
Residual cost	Storage, transport, disposal or further processing of every output/reject.	Receiver documents and invoices.
Customer acceptance	Structured sponsor/operator assessment against pre-agreed criteria.	Signed evaluation form.

Every KPI needs a unit, boundary, source, owner, frequency, missing-data rule and acceptance criterion. A dashboard without these controls can look precise while remaining non-defensible.

7. Site readiness and installation release

Area	Release question
Physical fit	Are access, footprint, loads, clearances, lifting and service zones confirmed?
Utilities	Are electrical, water, drainage, ventilation and communication interfaces verified?
Safety	Are risk assessment, guarding, emergency response, lockout and training complete?
Environmental	Are storage, spills, wastewater, odor, pests and contingency routes controlled?
Feedstock	Are receipt, inspection, rejection, segregation and maximum holding-time rules active?
Output	Is pending material quarantined and is a lawful off-spec/contingency receiver contracted?
Data	Are meters, calibration, access, backups, time synchronization and retention ready?
Commercial	Are insurance, ownership, damage, confidentiality and removal terms signed?

NO OUTPUT-ROUTE IMPROVISATION

Do not install a machine while assuming that a useful or legal destination for the resulting material will be found later.

8. Contract and data rights

Clause	What it should resolve
Purpose and scope	Defined experiment, configuration, materials, location and term.
Ownership and risk	Equipment title, custody, damage, insurance and loss.
Operating duties	Loading, cleaning, monitoring, maintenance, access and response.
Data	Ownership, access, raw-data retention, permitted analysis and audit rights.
Confidentiality	Protected information, exclusions, duration and permitted disclosures.
Publication	Review period, anonymization, approved claims, names, images and quotations.
Output and waste	Legal holder, testing, storage, transfer, off-spec and final disposition.
Stop/removal	Safety stop, breach, technical failure, end-of-term and site restoration.
Commercial option	Purchase or extension mechanics without implying guaranteed conversion.

The signed agreement should be reviewed by qualified counsel for the site and transaction. This framework identifies commercial controls; it is not contract language or legal advice.

9. Operating window and evidence integrity

Control	Required practice
Commissioning separation	Do not mix training, tuning and troubleshooting data into the accepted evidence window.
Configuration freeze	Record hardware, software, settings and approved changes; define when retesting is required.
Representative window	Cover normal variability and enough accepted batches to support the stated conclusion.
Deviations	Record missing data, unusual feedstock, interruptions and customer-side changes in real time.
Failure coding	Separate planned maintenance, customer causes, utility loss, process alarms and machine failures.
Raw records	Retain original data; corrections are traceable and never overwrite the source.
Independent evidence	Use competent laboratories or verification where a claim or decision needs added confidence.

TRUTH BEFORE OPTICS

A failure that is correctly recorded and resolved can strengthen the evidence package. An unexplained exclusion weakens the entire pilot.

10. Customer economics

The pilot model should compare the customer's current annualized cost with the annualized cost of the validated Rapid Organic use case. It must include displaced and newly created costs.

Current-state cost	Pilot/future-state cost
Waste storage, internal handling and cleaning	Equipment capital or lease and financing
Hauling, disposal, surcharges and contract minimums	Electricity, water, consumables and labour
Refrigeration or special containment where documented	Planned service, repairs, spares and downtime
Documented incident, pest or nuisance cost	Residual/output testing, storage, transport and disposition
Space or operational constraint where defensible	Training, compliance, insurance and site changes
Only evidence-backed current liabilities	Conservative contingency and end-of-life assumptions

ECONOMIC RULE

Assign zero revenue to the output until a lawful, repeatable receiver and evidence-supported net value are demonstrated. Avoided disposal cost and product revenue are separate claims.

11. Sensitivity and decision thresholds

Variable	Why it matters	Required treatment
Annual feedstock mass	Determines utilization and avoided cost.	Use measured range and downside case.
Hauling/disposal rate	Often varies by contract and geography.	Use actual terms; test escalation and minimum charges.
Availability	Reduces processed quantity and can add service cost.	Use accepted-window result and conservative future assumption.
Energy and labour	Can erase apparent savings at small scale.	Meter and time; include all incremental work.
Residual fraction	Creates handling and disposal obligations.	Measure every stream and include contingency route.
Output value	Often uncertain before regulatory and market proof.	Base case zero; upside only with evidence.
Capital/service price	Controls payback and customer approval.	Show purchase/lease scenarios and risk allowance.

A credible result states the range over which the decision remains positive. One attractive point estimate is not enough for a capital purchase decision.

12. Stop, correct, extend or continue

Condition	Typical decision
Safety control fails or legal route is lost	Stop immediately; quarantine material and investigate.
Machine or feedstock leaves approved configuration	Pause; assess deviation and define retest scope.
Data completeness falls below approved threshold	Extend only if the gap can be corrected without bias.
Customer operation materially changes baseline	Re-baseline or report a non-comparable result.
KPI misses but root cause is understood and correctable	Revise under change control; do not rewrite original criteria.
Economics fail under realistic assumptions	Stop, redesign or narrow the segment; do not manufacture savings.
Criteria pass with limitations	Proceed to evaluation and state the limitations explicitly.

PRE-AUTHORIZED HONESTY

Stop rules protect both parties. They prevent pressure to continue an unsafe, unlawful or commercially meaningless demonstration merely because it is public.

13. Final evaluation and customer acceptance

Final-report section	Minimum content
Question and scope	Decision question, site, dates, configuration, feedstock and exclusions.
Baseline	Sources, period, calculations, confidence and customer approval.
Operation	Accepted runs, quantities, availability, interventions, failures and deviations.
Output	Characterization, storage, status, receiver and off-spec disposition.
Economics	Measured costs, annualization, sensitivity and unmonetized effects.
Customer view	Sponsor and operator evaluation, training burden, concerns and continuation intent.
Limitations	What the pilot did not test or prove.
Decision	Buy, extend, repeat, revise or stop; owners and next actions.

Customer signature should confirm the baseline, factual site data and decision—not force endorsement of marketing language.

14. Case-study release and claims

Claim level	Permitted approach
Observation	State what happened at the named or anonymized site with dates and conditions.
Measured result	Give metric, method, denominator, accepted window and limitations.
Customer quotation	Use only written, approved wording with clear attribution.
Economic result	Separate measured pilot economics from projected annual scenarios.
Environmental statement	Use measured quantities and bounded comparisons; avoid unsupported impact claims.
Product/output statement	Follow WP-005; do not imply fertilizer, compost or sale status without support.
Broader market claim	Require replicated evidence or clearly label the conclusion as a hypothesis.

PUBLICATION TEST

A public statement should remain accurate if the reader sees the methods, rejected runs, uncertainty and limitations—not only the headline number.

15. From one pilot to repeatable sales

Post-pilot asset	Why it matters
Use-case specification	Defines the customer, feedstock, site and value conditions that produced success.
Validated configuration	Prevents sales from promising untested hardware, software or capacity.
Site-readiness checklist	Reduces installation surprises and qualification cost.
Commercial calculator	Reuses approved baseline logic and sensitivity structure.
SOP and training package	Converts operator learning into repeatable deployment.
Service and spares model	Prices real maintenance and response needs.
Evidence-backed claims register	Controls what sales and marketing may state.
Replication plan	Specifies the next sites needed to test geography, season and customer variability.

The acquisition value of a flagship pilot comes from reusable commercial assets and disciplined evidence, not from the logo of the host customer alone.

16. Frequently asked questions

Should the first pilot be free?

Usually not. The customer should contribute money, labour, site resources, data or another documented commitment. A no-cost demonstration can hide weak demand and unclear responsibilities.

How long should the pilot run?

Long enough to separate commissioning from a representative operating window and capture meaningful variability. The decision should be evidence-based, not tied to an arbitrary marketing date.

Can the pilot use several waste streams?

The core evidence window should use one approved envelope. Additional materials may be explored separately under controlled protocols.

What if the machine performs well but the ROI is weak?

That is a valid result. Reprice, redesign, target a different segment or stop; do not treat technical success as commercial validation.

Does a customer purchase prove product-market fit?

It is strong evidence for that use case, but repeatability still requires additional sales, delivery, service and retention evidence.

Can Rapid Organic publish all collected data?

Only according to the signed data and publication terms, with customer review and appropriate protection of confidential information.

What is the next practical action?

Select two or three candidate sites, apply the CP0 scorecard, and develop a baseline discovery package only for the strongest qualified candidate.

17. Conclusion

A flagship pilot should create a decision, not merely a demonstration. The strongest pilot begins with a costly customer problem, locks the comparison before operation, controls technical and legal boundaries, records failures as carefully as successes, and ends with a documented commercial choice.

RECOMMENDED NEXT MILESTONE

Rapid Organic should complete CP0 qualification for a short list of Canadian sites and advance only the candidate with the best combination of feedstock fit, baseline quality, sponsor commitment, lawful output route and credible purchase path.

Related Rapid Organic documents

- [WP-003 — How to Build a Defensible Organic-Waste Business Case](#)
- [WP-004 — Technical Validation Framework](#)
- [WP-005 — Output Characterization and Regulatory Pathways in Canada](#)
- [Rapid Organic Public Knowledge Base](#)

Document control

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